

Employee Attrition Prediction – Executive Summary

This project analyzed employee data to identify the main factors associated with employees leaving the company. By studying information such as job role, department, monthly income, work-life balance, years at the company, and job satisfaction, the analysis highlighted patterns that can help HR identify employees who may be at a higher risk of resigning.

The results indicate that employee attrition is influenced by several factors rather than salary alone. Employees with lower job satisfaction, fewer years at the company, limited work-life balance, and certain job roles showed a greater likelihood of leaving. Departments and roles with higher attrition rates should be prioritized for retention efforts, especially employees who are in the early years of their employment.

The predictive model developed in this project can help HR identify employees who may require additional support before they decide to leave. This enables the company to take proactive actions such as career development discussions, mentoring, workload adjustments, and employee engagement initiatives instead of reacting after resignations occur.

Based on these findings, it is recommended that HR conduct regular employee satisfaction surveys, strengthen career growth opportunities, improve work-life balance initiatives, and hold retention discussions with employees identified as being at higher risk. While the prediction system provides valuable guidance, it should be used as a decision-support tool rather than the sole basis for HR decisions, as personal circumstances and future workplace changes cannot be fully captured by historical employee data.

Overall, this project demonstrates how data-driven insights can help organizations reduce employee turnover, improve workforce stability, and support better strategic HR planning.